Predicting Stellar Labels for the Sagittarius Stream Using Gaia DR3 XP Spectra

Giovanny Rosales, Hunter Morgan

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Abstract

High-resolution spectroscopic surveys provide precise stellar parameters and chemical abundances, but cover only a fraction of the stars observed by Gaia. This project aims to transfer stellar labels from APOGEE DR17 to Gaia DR3 BP/RP XP spectra using a multi-output neural network and then apply the trained model to candidate members of the Sagittarius stellar stream. The target labels are effective temperature, surface gravity, metallicity, nitrogen-to-oxygen abundance, and aluminum-to-iron abundance.

This project stage extends the completed Sagittarius data-acquisition pipeline with a runnable workflow for constructing the APOGEE DR17–Gaia DR3 XP training sample. The workflow loads the APOGEE DR17 `allStarLite` catalogue, applies explicit stellar-label quality cuts, derives [N/O] from calibrated abundance columns, verifies Gaia source identifiers, and selects stars with XP continuous spectra. It records the number of rows remaining after each cut and supports restart-safe XP downloads. The previously constructed Sagittarius application sample contains 55,192 catalogue records, 42,386 unambiguous Gaia DR3 matches, and 27,069 sources with XP continuous spectra. At the latest verified checkpoint, 25,850 Sagittarius XP sources had been downloaded. The APOGEE workflow loaded 733,901 rows, verified 448,728 unique source identifiers in Gaia DR3, found XP continuous spectra available for 405,160 sources, and retained 334,486 sources after target-related warning filtering. The coefficient download has not yet begun because a test request encountered a temporary Gaia DataLink HTTP 503 error; consequently, model results are not reported in this checkpoint.

1 Introduction

The chemical and dynamical properties of stars preserve information about the assembly of the Milky Way and its accreted satellites. Metallicity and elemental-abundance ratios can distinguish stellar populations, constrain enrichment histories, and identify debris associated with disrupted dwarf galaxies. The Sagittarius dwarf spheroidal galaxy and its tidal stream provide an important setting for studying chemical evolution during an ongoing accretion event.

APOGEE DR17 supplies near-infrared stellar parameters and abundances for a large spectroscopic sample [1]. Gaia DR3 provides BP/RP low-resolution spectrophotometry, commonly called XP spectra, for a much larger population [3, 2]. Previous work demonstrates that data-driven methods can infer stellar parameters and selected abundances from Gaia XP data [7, 8, 5].

The project goals are to:

1. construct an APOGEE DR17 and Gaia DR3 XP training sample;
2. predict $T_{\rm eff}$, $\log g$, [Fe/H], [N/O], and [Al/Fe] from normalized XP coefficients;
3. evaluate predictive accuracy on held-out stars; and

4. apply the validated model to Sagittarius stream candidates with XP spectra.

2 Data

2.1 Gaia DR3 XP Spectra

Gaia DR3 publishes BP and RP mean spectra using coefficients that multiply internally calibrated basis functions [2]. The model input uses 55 BP and 55 RP coefficients. Each group is divided by its first coefficient, and the resulting unit-valued first coefficient is removed from each group. Therefore,

$$N_{\mathrm{XP}} = (55 - 1) + (55 - 1) = 108 \tag{1}$$

features remain per source. Sources with missing or nonfinite coefficients, an incorrect coefficient count, or a zero first coefficient will be rejected.

2.2 APOGEE DR17 Labels

The training labels come from the APOGEE DR17 `allStarLite-dr17-synspec_rev1.fits` catalogue [1]. The workflow uses `TEFF`, `LOGG`, `FE_H`, `N_FE`, `O_FE`, and `AL_FE`. The nitrogen-to-oxygen label is derived as

$$[\mathrm{N/O}] = [\mathrm{N/Fe}] - [\mathrm{O/Fe}], \tag{2}$$

so the target vector is

$$\mathbf{y}_i = (T_{\mathrm{eff}}, \log g, [\mathrm{Fe/H}], [\mathrm{N/O}], [\mathrm{Al/Fe}])_i \,. \tag{3}$$

The workflow also reads `SNR`, `ASPCAPFLAG`, `STARFLAG`, coordinates, `APOGEE_ID`, and the available Gaia source-identifier field.

2.3 Sagittarius Application Sample

The application sample comes from the 55,192-source Sagittarius catalogue of Vasiliev et al. [10]. Since this catalogue contains Gaia DR2 identifiers, the completed PS4 pipeline joins them through `gaiadr3.dr2_neighbourhood` and `gaiadr3.gaia_source`. This application sample is kept separate from the labeled APOGEE training sample.

3 Implemented PS5 Training-Set Workflow

The notebook `src/build_training_set.ipynb` is organized into small, testable sections:

1. configure paths, thresholds, and batch size;
2. download or load the APOGEE DR17 `allStarLite` FITS file;
3. verify that all required columns are present;
4. construct [N/O] from `N_FE-O_FE`;
5. apply sequential quality cuts and save a cutflow table;
6. obtain and validate Gaia source identifiers;
7. join against `gaiadr3.gaia_source` and require `has_xp_continuous=True`;

8. save the cleaned labels and identifiers; and

9. optionally download XP continuous products in restart-safe batches.

The quality policy rejects sources with the `STAR_BAD` bit set in `ASPCAPFLAG`, requires `SNR`≥ 70, requires finite target labels, and retains stars within $3500 \leq T_{\rm eff} \leq 7000$ K and $0 \leq \log g \leq 5$. After XP availability was established, sources carrying target-related `TEFF_WARN`, `LOGG_WARN`, `M_H_WARN`, or `N_M_WARN` bits were removed. The `VMICRO_WARN` bit was not used as an exclusion because microturbulence is not a prediction target and applying that warning would remove a substantial additional fraction of the sample. The loaded `allStarLite` product contained `PARAMFLAG` but not `ELEMFLAG`; therefore, `N_M_WARN` is used as the available conservative nitrogen-related warning rather than guessing an unavailable element-flag index.

The crossmatch uses the APOGEE-provided Gaia source identifier and verifies it directly in Gaia DR3. A single uploaded-table query initially returned exactly 50,000 rows, indicating truncation. The workflow was therefore revised to submit unique identifiers in restart-safe batches of 40,000, save each completed query batch, combine the results, and check for duplicates and unreturned identifiers. All 448,728 submitted identifiers were verified, none were unreturned, and no duplicates remained.

4 Observed Data-Processing Results

4.1 Sagittarius Pipeline

Table 1 reports the verified PS4 application-sample results.

Table 1: Verified Sagittarius catalogue and Gaia DR3 XP results.

Measurement	Count
Sagittarius catalogue sources	55,192
DR2 sources with at least one DR3 candidate	46,014
DR2 sources with exactly one DR3 candidate	42,386
DR2 sources with multiple DR3 candidates	3,628
DR2 sources without a returned DR3 candidate	9,178
Unique DR3 matches with XP continuous spectra	27,069
XP availability among unique matches	63.86%
XP sources downloaded at latest checkpoint	25,850
XP sources pending at latest checkpoint	1,219
Download completeness	95.50%

4.2 APOGEE Training-Sample Cutflow

Table 2 reports the measured reduction from the original APOGEE catalogue to the final quality-filtered metadata sample. All values were produced by `src/build_training_set.ipynb`.

All 448,728 submitted source identifiers were returned by Gaia DR3. XP continuous spectra were available for 405,160 sources, or 90.29% of the verified sample. Removing target-related warnings retained 334,486 sources, or 82.56% of the XP-available sample. The final metadata table contains 334,486 unique source identifiers and zero duplicates.

Table 2: APOGEE DR17–Gaia DR3 XP training-sample cutflow.

Selection	Before	After	Removed
Reject `ASPCAPFLAG STAR_BAD`	733,901	691,591	42,310
Require `SNR`≥ 70	691,591	597,478	94,113
Require five finite targets	597,478	495,611	101,867
Require $3500 \leq T_{\rm eff} \leq 7000$ K	495,611	495,611	0
Require $0 \leq \log g \leq 5$	495,611	495,464	147
Require valid unique Gaia source ID	495,464	448,728	46,736
Verify identifier in Gaia DR3	448,728	448,728	0
Require XP continuous availability	448,728	405,160	43,568
Reject target-related ASPCAP warnings	405,160	334,486	70,674

4.3 ASPCAP Warning Assessment

Before the final target-warning cut, the 405,160-source XP-available sample contained 11,958 `TEFF_WARN` flags (2.95%), 3,966 `LOGG_WARN` flags (0.98%), 150 `M_H_WARN` flags (0.04%), and 57,381 `N_M_WARN` flags (14.16%). Because these flags may overlap within a source, their individual counts do not sum to the 70,674 rows removed by the combined mask. For comparison, 92,018 sources (22.71%) carried `VMICRO_WARN`, 26,521 (6.55%) carried `STAR_WARN`, and none carried `STAR_BAD` after the earlier quality cut.

4.4 XP Coefficient Download Status

A 10-source Gaia DataLink test was attempted using `retrieval_type=''XP_CONTINUOUS''` and `data_structure=''DATAMODEL_GAIA''`. The service returned HTTP 503 Service Unavailable, indicating temporary maintenance or capacity limitations. The metadata construction, Gaia verification, XP-availability check, and quality filtering were already complete and were not affected. No APOGEE XP coefficient products are claimed as downloaded in this checkpoint. The downloader is batched and restart-safe so retrieval can resume without repeating completed batches when DataLink becomes available.

4.5 Target-Label Inspection

The XP-available metadata sample was inspected using minima, maxima, means, standard deviations, and the 1st, 5th, 50th, 95th, and 99th percentiles for all five labels. The observed temperature and gravity values remained within the configured ranges. The most extreme abundance values were flagged for row-level inspection before model fitting. Final distribution plots will be regenerated from the 334,486-source warning-filtered sample after the XP coefficients are retrieved so that the reported labels and model-ready features refer to the same sources.

5 Planned Preprocessing and Reproducible Splitting

For every retained XP source, BP and RP arrays will be checked for exactly 55 coefficients. Each array will be normalized independently by its first coefficient, and the first normalized value will be removed, producing 108 features. Source identifiers, features, and labels will be filtered using the same row mask so their alignment is preserved.

The cleaned sample will be divided using random seed 42 into 70% training, 15% validation, and 15% test subsets. Feature and target scalers will be fitted only on the training subset. The

source identifiers assigned to each split and the fitted preprocessing objects will be saved so the experiment and Sagittarius application can be reproduced exactly. The observed split counts will be reported after this stage is run.

6 Planned Baseline Model and Evaluation

The baseline is a PyTorch multi-output multilayer perceptron [9] with 108 inputs, two 128-unit ReLU hidden layers, dropout probability 0.2 after each hidden layer, and five regression outputs. Training will use validation-based early stopping. The selected checkpoint and preprocessing objects will be saved together.

Evaluation will report mean absolute error and root mean squared error separately for each target on the validation and test subsets. Training and validation loss curves, predicted-versus-true plots, and residual plots will be saved. Metrics will be inserted only after the model has been trained and evaluated on the fixed splits.

No validation or test metrics are reported for PS5 because the Gaia DataLink outage prevented retrieval of the model inputs. Reporting placeholder or smoke-test metrics would not represent held-out model performance.

A later refinement may predict a mean and variance for each target using a heteroscedastic Gaussian negative log-likelihood. Monte Carlo dropout may then be used to estimate epistemic uncertainty [4, 6]. This refinement will not replace the deterministic baseline until baseline performance has been established.

7 Limitations and Next Steps

The labeled metadata sample is complete, but verified XP availability does not mean that the coefficient arrays have been downloaded. The current external limitation is the HTTP 503 response from Gaia DataLink. In addition, `ELEMFLAG` is absent from the selected `allStarLite` product, so the nitrogen-related warning policy relies on the available `N_M_WARN` bit. The final abundance distributions and extreme values will be reviewed again after coefficient retrieval and row alignment.

Before model training, the remaining preparation tasks are:

1. download XP coefficients for the 334,486 quality-filtered sources in resumable batches;
2. combine the batches and record downloaded, missing, unexpected, and duplicate identifiers;
3. verify that every retained BP and RP array contains exactly 55 coefficients;
4. implement the 108-feature normalization and preserve identifier-label alignment;
5. create fixed 70%/15%/15% training, validation, and test splits with random seed 42; and
6. fit and save feature and target scalers using only the training subset.

The baseline PyTorch model, held-out metrics, diagnostic plots, and preliminary Sagittarius predictions are reserved for the next project stage after these preparation steps are complete.

8 Conclusion

PS5 produced a reproducible APOGEE DR17–Gaia DR3 training-metadata workflow in addition to the completed Sagittarius application pipeline. Starting from 733,901 APOGEE rows, the workflow applied documented label and quality requirements, verified all 448,728 retained unique identifiers in Gaia DR3, identified 405,160 sources with XP continuous spectra, and retained 334,486 unique sources after target-related warning filtering. The attempted coefficient test was blocked by a temporary Gaia DataLink HTTP 503 response, so coefficient validation and model training are not claimed as completed. The saved cutflow, metadata, chunked queries, and restart-safe downloader allow the remaining preparation to continue without repeating the completed crossmatch work.

Code and Data Availability

The implementation and reproduction instructions are available at https://github.com/LightsRedemption/DeepLearning_TeamAstro. Large downloaded and generated FITS products are retained locally and excluded from normal Git tracking because they can be regenerated from the documented pipeline.

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